

DSA Group Role-Play Report

Topic 5: Tree Traversals — Pre-order, In-order and Post-order

Group Members

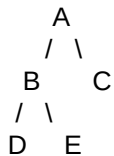
Name	ID Number
Fanuel Aregay	CEP0053/17
Abrham Nega	CEP0008/17
Aman Haftu	CEP0015/17
Habtamu Gizaw	CEP0064/17
Fikremaryam Belete	CEP0056/17

1. Chosen Topic and Rationale

Our group selected Tree Traversals from Chapter 5. Physical role-play makes the visiting order of a tree visible and easier to follow. Students act as nodes while the narrator announces the traversal rule and each visiting step.

2. Initial Tree Structure

Tree used in the simulation:



3. Traversal Rules

Traversal	Rule	Result
Pre-order	Root → Left → Right	A → B → D → E → C
In-order	Left → Root → Right	D → B → E → A → C
Post-order	Left → Right → Root	D → E → B → C → A

4. Step-by-Step Operation Log

Traversal	Step	Visited node
Pre-order	1	A
Pre-order	2	B
Pre-order	3	D
Pre-order	4	E
Pre-order	5	C
In-order	1	D
In-order	2	B
In-order	3	E
In-order	4	A
In-order	5	C

Traversal	Step	Visited node
Post-order	1	D
Post-order	2	E
Post-order	3	B
Post-order	4	C
Post-order	5	A

5. Operation Tally

Pre-order: 5 visited nodes (A, B, D, E, C)

In-order: 5 visited nodes (D, B, E, A, C)

Post-order: 5 visited nodes (D, E, B, C, A)

6. Critical Reflection: Limitations

- Remembering the exact traversal rule while moving can be difficult.
- Returning from a child node to its parent is harder to demonstrate physically than in recursive code.
- Participants can accidentally move to the wrong node.
- Limited physical space can make the tree structure less clear.
- Human role-play does not directly show the recursive function-call stack.

7. Simulation Video

Working video link:

https://drive.google.com/file/d/1Qpbp3_CSUuEIMc50cF2BCIpESsm1HC-e/view?usp=drivesdk

8. Conclusion

The simulation demonstrates that Pre-order, In-order and Post-order use the same tree but apply different visiting rules. The role-play helped the group visualize the sequence of visits and understand tree traversal step by step.